

Addressing Design Requirements

The final prototype testing plan will address **all** of the high-priority design requirements, which are:

- **Durability/Longevity:** The device should be able to last several repeated uses including rough use and being dropped.
- **Accuracy:** The device must have a ± 1 mmHg accuracy in measuring maximal inspiratory pressures (MIP) and sniff inspiratory pressure (SNIP).
- **Simplicity and Replicability:** The device should be easy to assemble and replicate with minimal technical expertise and as little explanation as possible.
- **Affordability:** The devices must be able to be produced within our budget with minimal outside funding from Drexel/other parties.
- **Portability:** The devices should be easy to transport to and from Drexel and to transport from a closet to the instructional room. Each device should also be able to be carried by a college student.

Conduct of Tests

The testing plan will provide effective details for conducting physical and mathematical tests for each high-priority design requirement. Below are the specific tests:

1. Durability/Longevity Test

- **Setup:** The prototype will be tested several times including a rough test where we will squeeze it and hit it against objects. It will also be dropped from a height of 6 feet onto the floor.
- **Expected Results:** Device should maintain functionality without significant wear or failure over the course of the testing.
- **Minimum Result:** The prototype should remain fully functional after the testing.
- **Goal:** Ensure the device can perform as expected if the entire class of 80-100 uses it.

2. Accuracy Test

- **Setup:** The MIP will be hooked up to a syringe or other device that can reliably create a certain pressure vacuum to ensure we get the same reading every time.
- **Expected Results:** The measured MIP readings should fall within the expected range.
- **Minimum Result:** The device should fall within 2 mmHg.
- **Goal:** Achieve ± 0.5 mmHg accuracy, ensuring more precise measurements.

3. Simplicity and Replicability Test

- **Setup:** The prototype will be given to several teachers, students, and others with no instructions followed by another round of testing with instructions.
- **Expected Results:** Assembly should take no more than 10 complete with or without instructions.
- **Minimum Result:** The assembly should take no more than 30 seconds to complete with or without instructions.

- **Goal:** Ensure that the average person can assemble the device without issue.

4. Affordability Test

- **Setup:** There is no way to test affordability properly, so the only way to do so is by making it as cost effective as possible.

5. Portability Test

- **Setup:** The prototype will be tested for weight, ease of handling, and storage by being carried by different individuals and placed in typical storage conditions.

Instead of testing the portability, which will be almost impossible to do, we will focus on building secure packaging, ensuring that our prototype does not break. It will be in a hard casing.
- **Expected Results:** The device should weigh under 2 kg and be easy to carry and store in typical classroom spaces. The device should also be able to fit into our specific casing.
- **Minimum Result:** The weight should not exceed 2 kg, and the device should be transportable and storable.
- **Goal:** Target weight under 1 kg to ensure maximum portability.

Each test will involve **3 or more trials** with 2 measurements each to reduce random error, ensuring the results are statistically reliable.

Logic of the Plan

Each test is designed to evaluate the ability of the prototype to meet specific design requirements. For example:

- **Durability:** Repeated use tests will simulate classroom conditions to ensure the device can withstand wear over time.
- **Accuracy:** Comparing MIP readings to a calibrated reference ensures precise measurements.
- **Simplicity and Replicability:** Assembly tests will confirm that the device can be easily constructed by multiple individuals.
- **Affordability:** Cost estimation tests ensure that the device can be produced at an affordable price for mass distribution.
- **Portability:** The weight and size testing ensures that the device is easy to transport and store.

These tests will follow scientific best practices, including controlling variables, setting clear success thresholds, and performing multiple trials to ensure validity.

Use of Field Experts

We will consult **two field experts** to confirm the validity of the testing plan and incorporate their feedback:

1. **Dr. Joseph Sarver** (Biomechanics and mechanical testing expert)
2. **Jaimie Dougherty** (Biomedical engineering and functional testing expert)

Both experts will review the testing procedures and provide suggestions for improvements. Their feedback will be documented, and any necessary adjustments will be made to the plan based on their advice.

This comprehensive testing plan will ensure that the MIP device meets all high-priority design requirements, and we will document all the results to validate the design's effectiveness.